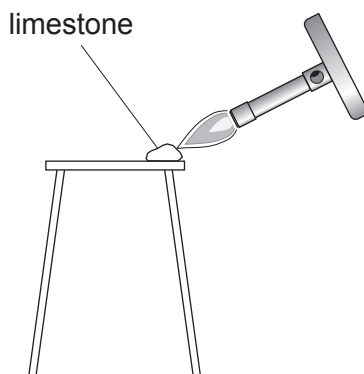
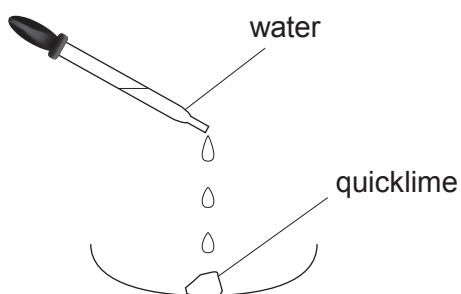


5. (a) A student carried out a two-stage experiment to change limestone (calcium carbonate) into slaked lime (calcium hydroxide).

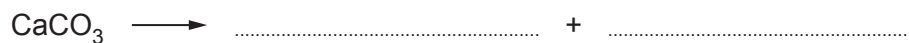


Stage 1: Limestone (calcium carbonate) decomposes into quicklime (calcium oxide) and carbon dioxide



Stage 2: Quicklime (calcium oxide) reacts with water forming slaked lime (calcium hydroxide)

- (i) Write the formulae for calcium oxide and carbon dioxide to complete the equation for the reaction taking place in stage 1. [2]



- (ii) Calcium hydroxide contains one Ca^{2+} ion for every two OH^- ions.

Write the chemical formula for calcium hydroxide. [1]

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